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E350

U3400
RELEASE NOTES FOR F/W VERSION R4.6

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Version	Date	Comments
1.0	11/04/2016	Initial Version
1.1	15/07/2016	Clarification of firmware upgrade restriction

Important Information

This document provides information relating to E350 U3400 meter firmware (the "Product")

Landis & Gyr Pty Ltd (ABN 78 002 894 224) trading as Landis+Gyr.

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Support

For any further product questions and support, please contact sales.au@landisgyr.com or alternatively call 1 300 252 634. For international callers, contact number is +61 2 9690 7333.

Landis+Gyr
241 O'Riordan Street
Mascot NSW 2015
Australia
+61 2 9690 7333
sales.au@landisgyr.com
www.landisgyr.com

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1. About This Document

1.1. Purpose

The purpose of the Release Notes document is to communicate major new features and changes in this release of U3400 firmware. It also documents known problems and workarounds.

1.2. Scope

This document describes the E350 - U3400 firmware version R4.6.

1.3. Definitions, Acronyms, and Abbreviations

LC: Load Control
QoS: Quality of Supply
LP: Load Profile
AMI: Advanced Metering Infrastructure
SCC: Supply Capacity Control
RSM: Relay State Machine
PP: Partial Programming

1.4. References

E350 U3400 User Manual
EMPWin+ 2.9 Release notes

2. About the release, compatible products and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released Firmware	This Released	Compatible Meter	Upgradability	Downgradability
R4.5	**R4.6**	U3400	4.5 →4.6 With meter reprogram as noted below	4.6 →4.5 With meter reprogram

Broadly, this R4.6 firmware release provides following information:

(i) Gross Metering with flexible totalling on the phase registers

This firmware has been tested on the following platforms

- E350 U34xx meter
- EMPWin+ 2.9

Important Note: Due to the fundamental register totalling changes that this firmware introduces and meter external wiring required to take advantage of this functional change, this release of firmware is intended for new meters only. If customers intend to upgrade existing U3400 meters to this firmware for evaluation purposes, a full meter reprogramming is required.

3. New Features, Important Notes and Fixes

3.1. New Feature: Gross Metering flexible totalling

This R4.6 firmware allows customers to define what phase registers are used in the determination of the Total registers. Previously the Total registers in a polyphase meter were always A+B+C for import loads and another Total register for A+B+C exporting loads. These totals are available for kWh, kVARh and kVAh. Allowed options supported by EMPWin+ 2.9 are A+B+C, A+B, B+C or A+C.

Note that:

- F/W upgrading to R4.6 will not automatically change the method of calculation of the Total registers. The meter must be fully programmed with at least EMPWin+ 2.9 after firmware upgrade and if the method of Total calculation needs to be changed.
- F/W downgrade from R4.6 to an earlier version will cause the Total registers to accumulate as A+B+C again; existing Total values will not be lost. If downgrading F/W in a meter that has a two phase total program to R4.5 or earlier it is recommended to also fully program the meter.
- Full programming will be required to change the Total Metering option from the default of A+B+C.
- Full programming the meter with older programs that didn't support the new Total metering option will result in A+B+C totals.
- The Total Metering option selected will apply to any Total registers that are selected for Load Profiling.

3.2. Important Note: Display

When a Total register item is displayed the icons E1E2E3 are shown and they depict the Total metering option that is in use. If E1E2E3 are displayed then the Totals are based on A+B+C which is the traditional method of totalising in a polyphase meter. If Total method is A+B then just E1E2 will be

displayed, similarly if B+C is selected then E2E3 will be shown. In this way the meter programmer can always confirm the Total metering option that has been enabled in the meter (provided a Total register has been selected for display).

The Total Metering option selected will apply to the Instantaneous Total power items; Instantaneous Total power items are always calculated as Net. If A+B is selected and A is importing and B exporting a similar amount then the instantaneous total power could be zero regardless of the phase C load.

The power direction arrows also follow the Total Metering option. If A+B option is selected and A is importing and B is exporting then the icon arrow direction indicators might not display if A and B are of similar values (but different directions). Phase C is not considered in the arrow load direction.

3.3. Important Note: Firmware Upgrade

U3400 can be upgrade from R4.5 to R4.6 via the optical port but not via the remote port. Immediately after firmware upgrade the meter must be fully programmed.

3.4. Important Note: Firmware Downgrade (Rollback)

U3400 can be downgraded from R4.6 to R4.5. Immediately after firmware upgrade the meter must be fully programmed.

3.5. Important Note: Pulsed Outputs

The Total Metering option selected will apply to the pulsed outputs for any of the Total registers. They will always be Net even if Gross metering has been selected.

3.6. Important Note: Ripple Program

A ripple program should always been written to the meter at least once even if it is a no ripple program. This will ensure that the CRC check that is performed on power up for the ripple program area will not trigger a Non-volatile error warning (0x17).

3.7. Important Note: Supply Capacity Control

The Total Metering option selected will apply to the SCC calculation methods. For example if A+B is selected then SCC can only trigger if A+B exceeds the set limit; the size of the phase C load is not considered in SCC calculations.

3.8. Important Note: Test LEDS

The Total Metering option selected will apply to how the LEDs pulse. If A+B Totals have been programmed and A is importing and B exporting a similar amount then the LED will not pulse even if a heavy load is flowing in phase C.

3.9. Important Note: Billing Self Read Periods

The Total registers will now be based on the Total Metering method selected for both TOU and Non TOU Total registers.

3.10. Limitations:

- *Partial programming doesn't log the reference program used; this can be tracked via the software tool used if required.*

- *Programming log doesn't indicate the User ID, it is always 777.*

- *Group Broadcast Table write commands that are received by a meter with a non-matching address incorrectly count this as a Table Write error (they should just ignore the command).*

- *If two Breaker close commands are sent before the Breaker has had time to close then the AD will correctly operate but report 0's for the AD parameters.*

- *There could be a small current value displayed/reported even when no current is flowing in an element. The energy will not be creeping due to this minor problem.*

- *The display item Time Zone is incorrectly displayed if a negative time zone area has been selected e.g. USA time zones. The meter uses the correct TZ offset.*

- *The Alarm icon (bell) that is shown on the Display Test is not currently used.*

- *If the meter is fully programmed while the ARMing state timer is running, the ARMing state is correctly cancelled but an event is not created. This is seen as a minor issue because full programming is often followed by a manual request to clear the event log. Partial programming of the ARMing timers does correctly log an event if the timer is terminated.*

- *If outages are less than about 5 seconds duration then there is a possibility that the LC LOS events will be reported twice in the event log. Once before the actual Power Up event and once after this event. This is not a serious problem just duplicate event logging.*

- *The event associated with Breaker LOS operation (2126) is only being logged when the meter is powered down. On power up and the Breaker closes due to LOS timeout then it is expected that an additional event is logged but doesn't occur. This is not a serious problem just a missing event according to the technical specification.*

- *The triggered status of UFLC was not reset by the Toolbox command "Disable UFLC". UFLC was correctly disabled but the existing triggered state was not cleared. This will prevent UFLC from running if it is enabled again.*

- *If the Ripple Power Restore is set to 0 then after a power up the Ripple Time Table settings option will not be used to control the relays. The work around is to not use 0 for this random delay parameter.*

- *The RC icon on the display will slow flash if the meter contains a valid ripple program and it is in Failsafe mode. If the meter is reprogrammed to disable Ripple via the RSM settings then the slow flashing RC icon is expected to stop but it doesn't. It requires a power cycle to terminate the slow flashing of the RC icon.*

- *If the reconnect period is short and the Breaker reconnects within the same Trading period then the Breaker will disconnect at the next minute boundary, therefore short reconnect periods are not recommended.*